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UNIT - I

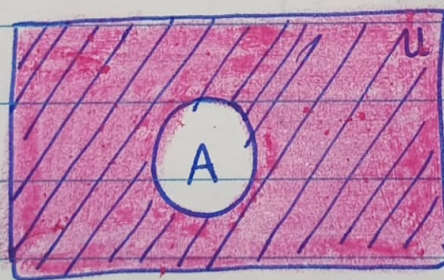
Sets

* Sets

Collection of well defined objects are called Sets and every member of it is called Element.

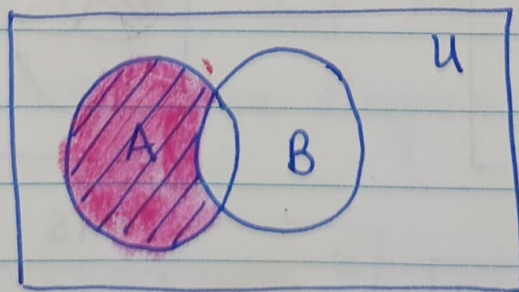
* Venn Diagram

* Compliment of set $A = A^c = A'$

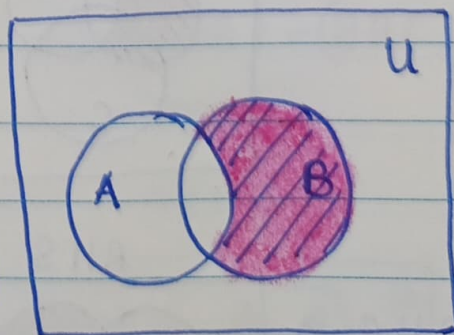


$$A' = u - A$$

* Difference of set A & B



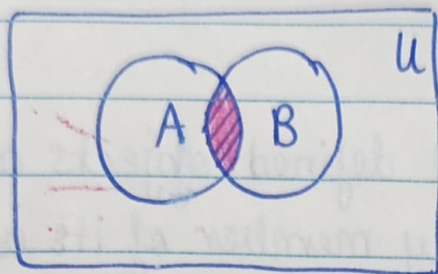
$$A - B$$



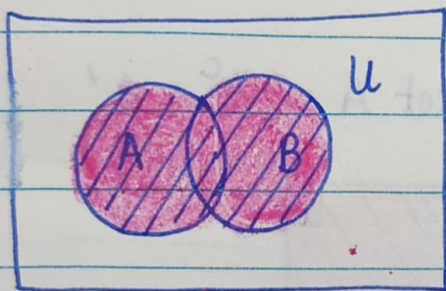
$$B - A$$

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* Intersection of A & B i.e $A \cap B$

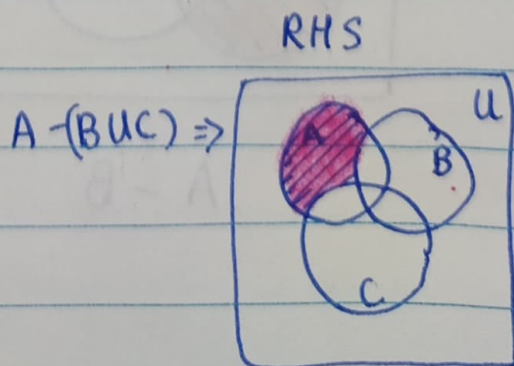
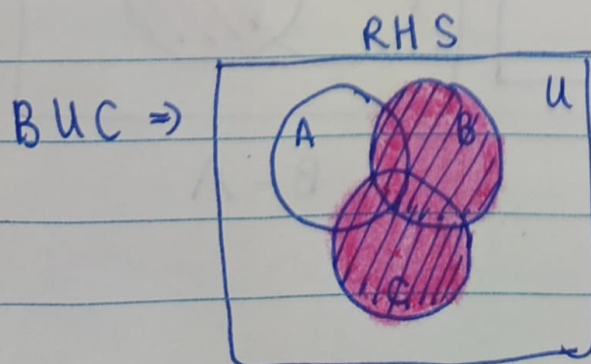
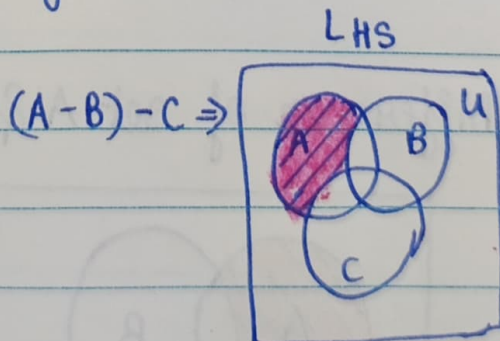
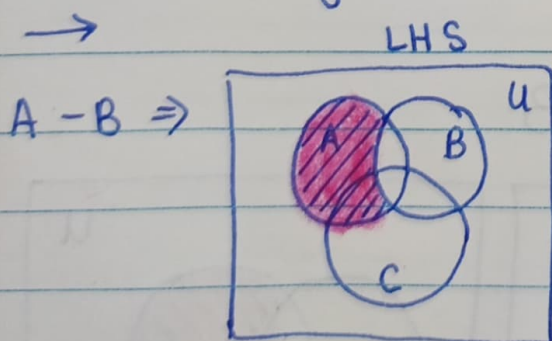


* Union of A & B i.e $A \cup B$



Example? Show that $(A - B) - C = A - (B \cup C)$
using venn diagram

→



$\therefore \text{LHS} = \text{RHS}$

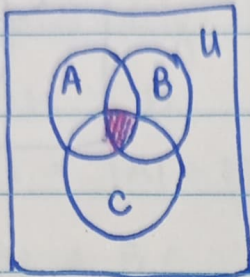
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Example > Show that $A \cap B \cap C = A - [(A - B) \cup (A - C)]$

→

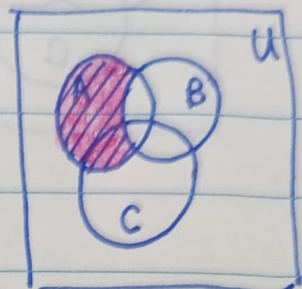
LHS

$$A \cap B \cap C =$$



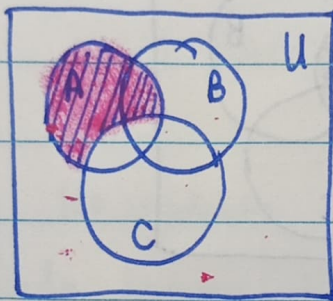
RHS

$$A - B =$$



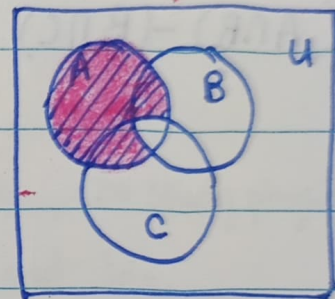
RHS

$$A - C =$$



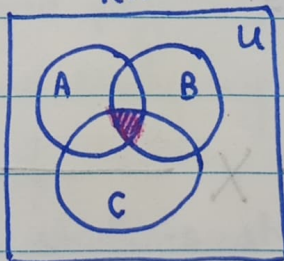
RHS

$$(A - B) \cup (A - C) =$$



RHS

$$A - [(A - B) \cup (A - C)] =$$



Thus,

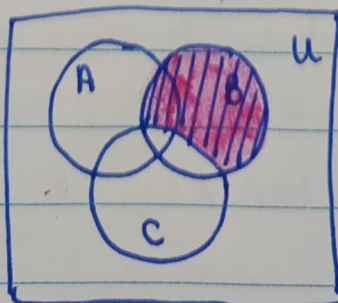
$$RHS = LHS$$

Example > Show that $A \cap (B - C) = (A \cap B) - (B \cap C)$ using venn diagram.

→

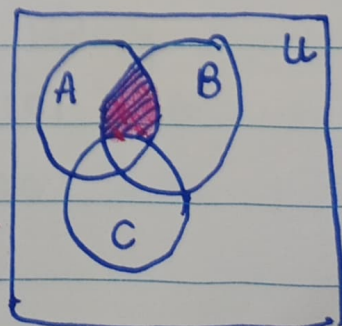
LHS

$$(B - C) \Rightarrow$$



LHS

$$A \cap (B - C) \Rightarrow$$

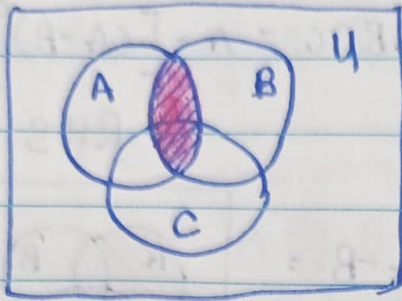


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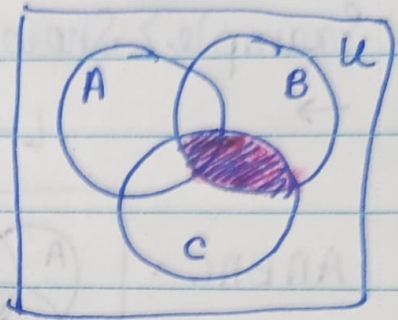
RHS

$A \cap B \Rightarrow$



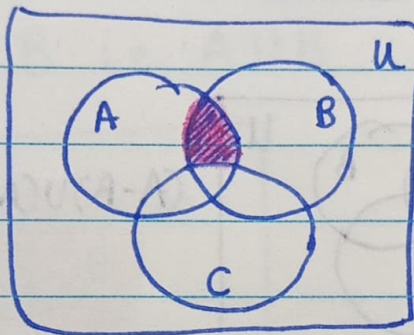
RHS

$B \cap C \Rightarrow$



RHS

$(A \cap B) - (B \cap C) \Rightarrow$



$\therefore \text{RHS} = \text{LHS}$

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* Principle of Inclusion & Exclusion

$|A|$ = cardinality of set A.

- For 2 sets A, B

$$|A \cup B| = |A| + |B| - |A \cap B|$$

- For 3 sets A, B, C

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Ex: Among 100 students, 32 study maths, 20 study physics, 45 study biology, 15 study maths and bio, 7 study maths and physics, 10 study physics and bio, 30 do not study any of the 3 subjects.

1) Find no. of students studying all 3 subj.

2) Find no. of students studying exactly 1 of 3 subj.

⇒ let A = set of students studying maths

B = set of students studying physics

C = set of students studying biology

$$|U| = 100, |A| = 32, |B| = 20, |C| = 45$$

$$|A \cap B| = 7, |A \cap C| = 15, |B \cap C| = 10$$

$$|\overline{A \cup B \cup C}| = 30.$$

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$$|A \cup B \cup C| = |U| - |\overline{A \cup B \cup C}|$$

$$|A \cup B \cup C| = 100 - 30$$

$$|A \cup B \cup C| = 70$$

By Inclusion - Exclusion principle.

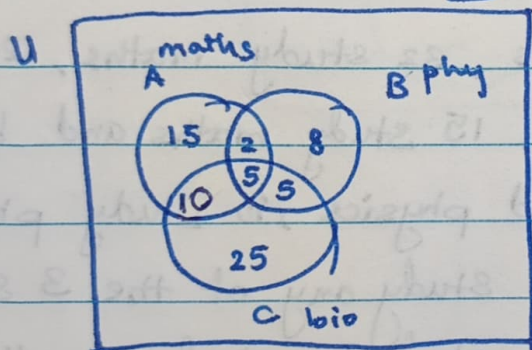
$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$70 = \frac{97}{52} \frac{32 + 20 + 45 - 7 - 15 - 10 + |A \cap B \cap C|}{-32}$$

$$70 = 97 - 32 + |A \cap B \cap C|$$

$$70 = 65 + |A \cap B \cap C|$$

$$|A \cap B \cap C| = 70 - 65 = \boxed{5}$$



⇒ no. of students studying all 3 subjects is 5.

no. of students studying exactly 1 subject is 48

Eg: It was found that in the 1st year of CS of 80 students.

50 students know OOP language, 55 know C language, and 46 know python. It was also known that 37 know C and OOP, 28 know C and Python, 25 know python & OOP, 7 know none of the language.

12 Find how many know all 3 languages.

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2x How many know exactly 2 languages.

$\Rightarrow A = \text{set of students who know OOP}$

$B = \text{set of students who know C}$

$C = \text{set of students who know python.}$

$$|U| = 80, |A| = 50, |B| = 55, |C| = 46$$

$$|A \cap B| = 37, |A \cap C| = 28, |B \cap C| = 25$$

$$|A \cup B \cup C| = 7.$$

By Inclusion-Exclusion Principle.

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$|A \cup B \cup C| = |U| - |\overline{A \cup B \cup C}|$$

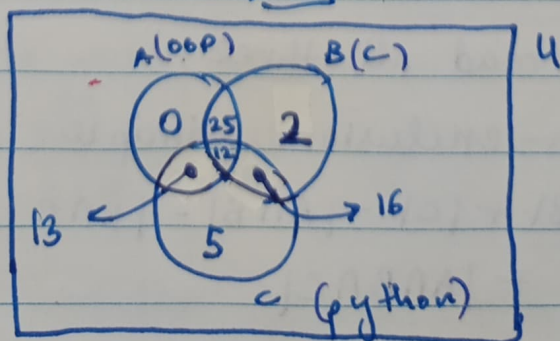
$$|A \cup B \cup C| = 80 - 7$$

$$|A \cup B \cup C| = 73$$

$$73 = \frac{50 + 55 + 46}{151} - \frac{37 + 28 + 25}{65} + x$$

$$73 = 61 + x$$

$$x = 73 - 61 = \boxed{12} = |A \cap B \cap C|$$



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no. of

⇒ Students who know all 3 languages is 12.

no. of students who know exactly 2 languages is 7.

Q In A Survey of 60 people, it was found that 25 read Business India, 26 read India Today, 26 read Times of India, 11 read both Business India & India Today, 9 read both Business India & Times of India, 8 read both India Today & Times of India. and 8 read none of these.

1. Find how many read all of these.

2. Find how many read exactly 1.

⇒ A = set of people who read Business India

B = set of people who read India Today

C = set of people who read Times of India

$$|A| = 25, |B| = 26, |C| = 11, |U| = 60$$

$$|A \cap B| = 11, |A \cap C| = 9, |B \cap C| = 8$$

$$|\overline{A \cup B \cup C}| = 8$$

(1) 8 people read all three.

using inclusion-exclusion principle

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

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$\Rightarrow |A \cup B \cup C| = 45 \Rightarrow$ people read at least 1 magazine.

remaining: $60 - 45 = 15 \rightarrow$ read none.

Exactly 1: $A \rightarrow |A| - (|A \cap B| + |A \cap C| - |A \cap B \cap C|) = 13$

$$B \rightarrow |B| - (|A \cap B| + |B \cap C| - |A \cap B \cap C|) = 15$$

$$C \rightarrow |C| - (|A \cap C| + |B \cap C| - |A \cap B \cap C|) = 17$$

Total who read exactly 1 = 45

Q7 A set of integers from 1 to 250. How many of these numbers are divisible by 3, 5, 7? Also indicate how many are divisible by 3 or 5 but not by 7.

$\Rightarrow A =$ set of integers divisible by 3

$B =$ set of integers divisible by 5

$C =$ set of integers divisible by 7.

$$|A| = \text{no. of integers divisible by 3} = \frac{250}{3} = 83.33 \approx 83$$

$$|B| = \text{no. of integers divisible by 5} = \frac{250}{5} = 50$$

$$|C| = \text{no. of integers divisible by 7} = \frac{250}{7} = 35.7 \approx 35$$

$$\begin{aligned} |A \cap B| &= \text{no. of integers divisible by 3 \& 5} \\ &= \frac{250}{3 \times 5} = 16.66 \approx 16.67 \approx \underline{16} \end{aligned}$$

* always consider 16 even when 16.8 / 16.9

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$|ANC|$ = no. of integers divisible by 3 & 7

$$= \frac{83.33}{3 \times 7} = \frac{83.33}{7} = 11.9 \approx 11$$

$|BNC|$ = no. of integers divisible by 5 & 7

$$= \frac{50}{5 \times 7} = \frac{50}{7} = 7.11 \approx 7$$

$|ANBNC|$ = no. of integers divisible by 3 & 5 & 7

$$= \frac{250}{3 \times 5 \times 7} = \frac{250}{105} = 2.37 \approx 2$$

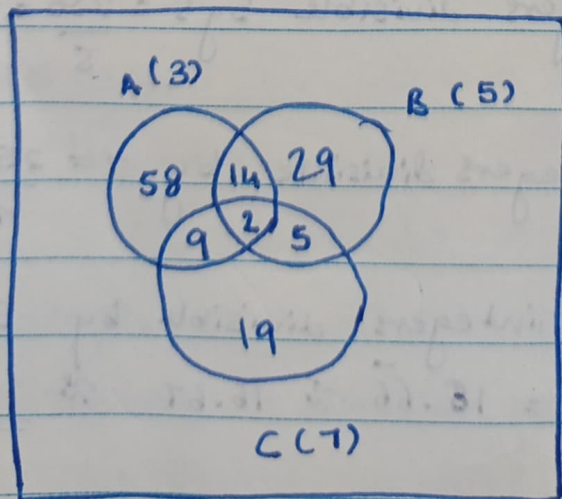
By Inclusion-Exclusion Principle.

$$|A \cup B \cup C| = |A| + |B| + |C| - |ANB| - |ANC| - |BNC| + |ANBNC|$$

$$|A \cup B \cup C| = 83 + 50 + 35 - 16 - 11 - 7 + 2$$

$$= 168 - 34 + 2 = 170 - 34$$

$$|A \cup B \cup C| = \boxed{136} = \text{no. of integers divisible by } 3/5/7.$$



Numbers divisible by 3, 5, 7 are $\boxed{136}$.

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no. of integers divisible by 3/5 not by 7

$$\Rightarrow 58 + 29 + 14 = \boxed{101}.$$

Q7 Among the integers 1 to 1000

1) How many of them are not divisible by 3 nor by 5 nor by 7.

2) How many are not divisible by 5 and 7. but divisible by 3.

$\Rightarrow A =$ set of integers divisible by 3

$B =$ set of integers divisible by 5

$C =$ set of integers divisible by 7.

$|A| =$ no. of integers divisible by 3

$$= \frac{1000}{3} = 333.\bar{3} \approx 333$$

$|B| =$ no. of integers divisible by 5

$$= \frac{1000}{5} = 200$$

$|C| =$ no. of integers divisible by 7

$$= \frac{1000}{7} = 142.8 \approx 142.$$

$|A \cap B| =$ no. of integers divisible by 3 and 5.

$$= \frac{1000}{3 \times 5} = \frac{333}{1} = 47$$

$|A \cap B|$
= no. of integers
divisible by

3 and 5

$$= \frac{1000}{3 \times 5} = 66.\bar{6}$$

$\approx 66.$

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$$|B \cap C| = \text{no. of integers divisible by 5 \& 7} = \frac{200}{5 \times 7} = \frac{200}{35} = 5.7 \approx 5$$

$$|A \cap B \cap C| = \text{no. of integers divisible by 3, 5 \& 7} = \frac{1000}{3 \times 5 \times 7} = \frac{1000}{105} = 9.5 \approx 9$$

By Inclusion-Exclusion Principle,

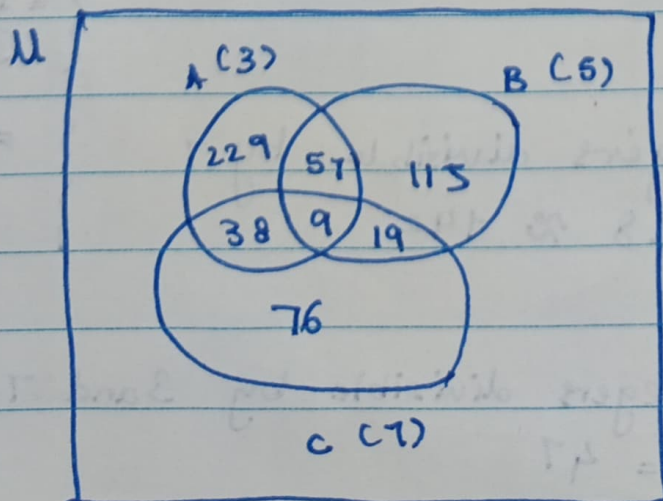
$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

$$|A \cup B \cup C| = 333 + 200 + 142 - 66 - 47 - 28 + 9 = 533 + 113 = 646$$

$$|A \cup B \cup C| = 684 - 141 = 543$$

$$|\overline{A \cup B \cup C}| = |U| - |A \cup B \cup C| = 1000 - 543 = 457$$

$|\overline{A \cup B \cup C}| = 457$ = no. of integers not divisible by 3 or 5 or 7.



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⇒ no. of integers not divisible by 3/5/7 is 451.

no. of integers divisible by 3 only is 229.

Qr A software company is testing an application for three major bugs: Bug A, Bug B, Bug C. Out of 100 test cases, 40 cases contain Bug A, 35 contain Bug B, 30 cases contain Bug C. Furthermore, 12 cases contain both A and B, 8 cases contain both B & C, 10 contain A & C. If 20 cases contain none, find:

- 1) No. of test cases containing all 3 bugs
- 2) No. of test cases containing exactly 2 bugs.

⇒ A = no. of cases containing Bug A.

B = no. of cases containing Bug B

C = no. of cases containing Bug C

$$|U| = 100$$

$$|A| = 40, |B| = 35, |C| = 30$$

$$|A \cap B| = 12, |B \cap C| = 8, |A \cap C| = 10$$

$$|\overline{A \cup B \cup C}| = 20$$

$$|A \cup B \cup C| = |U| - |\overline{A \cup B \cup C}| = 100 - 20 = 80$$

By Inclusion-Exclusion Principle,

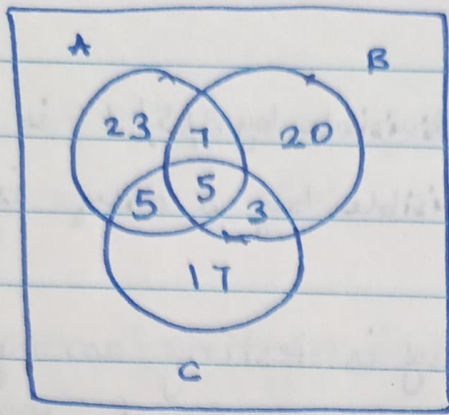
$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C|$$

$$80 = \underbrace{40 + 35 + 30}_{105} - \underbrace{12 - 8 - 10}_{-30} + x = \underbrace{105 - 30}_{75} + x$$

$$80 - 75 = x \Rightarrow |A \cap B \cap C| = \boxed{5}.$$

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— X —

$A = \text{no. of cases containing } A$
 $B = \text{no. of cases containing } B$
 $C = \text{no. of cases containing } C$

$$101 = 100$$

$$|A| = 40 \vee |B| = 35, |C| = 30$$

$$|A \cup B| = |A| + |B| - |A \cap B| = 40 + 35 - 10 = 65$$

$$|A \cup B \cup C| = 20$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$20 = 40 + 35 + 30 - 10 - 10 - 10 + x$$

$$20 = 115 - 30 + x$$

$$20 = 85 + x$$

$$x = 20 - 85 = -65$$

$$|S| = 100 \quad \therefore x = 25 - 80$$

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MATHEMATICAL INDUCTION

Mathematical Induction is an approach where we generalise a particular solution. If we have to prove a statement P , then,

1) Prove the statement P is true for $n=1$, called as BASIS STEP.

2) Assume statement P is true for $n=k$, called as HYPOTHESIS STEP.

3) Prove statement for $n=k+1$, called as INDUCTION STEP.

These are the three steps of mathematical Induction.

Eg: Using mathematical Induction, show that

$$P(n) : 1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $P(1) \quad P(2) \quad P(3)$

$\rightarrow (LHS = RHS)$

we have to generalise i.e. prove it is true.

$\Rightarrow$ Step 1 for $n=1$

$$LHS = P(1) = 1$$

$$RHS = \frac{P(1)}{2} = \frac{1(3 \cdot 1 - 1)}{2}$$

$$\text{as } LHS = RHS$$

$$RHS = P(1) = \frac{1(3(1)-1)}{2} = \frac{1(3-1)}{2}$$

$\therefore (P(1))$ is true.

$$\frac{2}{2} = \underline{\underline{1}}$$

Step 2 consider $P(n)$ is true for $n=k$.

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Replacing n by k ,

$$P(k) : 1 + 4 + 7 + \dots + (3k-2) = \frac{k(3k-1)}{2}$$

Step 3 :

we have to prove that $P(N)$ is true for $n=k+1$

(thus we write a statement for $n=k+1$, as well)

$$P(k+1) : 1 + 4 + 7 + \dots + \overbrace{(3k-2)}^{k^{\text{th}} \text{ term}} + \underbrace{(3(k+1)-2)}_{(k+1)^{\text{th}} \text{ term}}$$

add another term and replace k by $(k+1)$

$$= \frac{[k+1(3(k+1)-1)]}{2}$$

RHS

2

simplifying $\rightarrow$

$$\Rightarrow P(k+1) : 1 + 4 + 7 + \dots + (3k-2) + (3k+1) = \frac{(k+1)(3k+2)}{2}$$

considering LHS $\rightarrow$.

$$\text{LHS} = \underline{1 + 4 + 7 + \dots + (3k-2)} + (3k+1)$$

assuming the statement in step 2 is true we know that the underlined term simplifies to,

$$\Rightarrow \frac{k(3k-1)}{2} + (3k+1) = \text{LHS}$$

$$\Rightarrow \frac{k(3k-1)}{2} + 2(3k+1) = \frac{k(3k-1) + 6k + 2}{2}$$

$$\Rightarrow \frac{3k^2 - k + 6k + 2}{2} \Rightarrow \frac{3k^2 + 5k + 2}{2} \Rightarrow \frac{3k^2 + 3k + 2k + 2}{2}$$

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 $2 \mid 6$
 $3 \mid 3$
 1

$$\Rightarrow \frac{3k(k+1) + 2(k+1)}{2} \Rightarrow \frac{(k+1)(3k+2)}{2} = \text{RHS}$$

as LHS = RHS, hence proved.

Ex: Using mathematical Induction show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

$\Rightarrow$ Step 1 for $n=1$.

$$\text{LHS} = P(1) = \frac{1}{1 \cdot 2} = \frac{1}{2} \quad \text{RHS} = P(n) = \frac{n}{n+1}$$

$$\left. \begin{array}{l} \text{as LHS} = \text{RHS} \\ \therefore P(1) \text{ is true.} \end{array} \right\} \quad \text{RHS} = P(1) = \frac{1}{1+1} = \frac{1}{2}$$

Step 2 considering statement is true for $n=k$.

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$$

Step 3

we have to prove that statement is true for $n=k+1$.

LHS

$$P(k+1): \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+1+1)} = \frac{k+1}{k+1+1}$$

simplifying, we get $\rightarrow$

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$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

considering LHS $\Rightarrow$

assuming step 2 is true, we know that underlined term will simplify to

$$\Rightarrow \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \Rightarrow \frac{1}{k+1} \left(\frac{k+1}{1} \cdot \frac{1}{k+2} \right)$$

$$\Rightarrow \frac{1}{k+1} \left(\frac{k(k+2) + 1}{k+2} \right) \Rightarrow \frac{1}{k+1} \left(\frac{k^2 + 2k + 1}{k+2} \right)$$

$$\Rightarrow \left(\frac{k^2 + k + k + 1}{k+2} \right) \frac{1}{k+1} \Rightarrow \left(\frac{k(k+1) + 1(k+1)}{k+2} \right) \frac{1}{k+1}$$

$$\Rightarrow \left(\frac{(k+1)^2}{(k+2)} \right) \frac{1}{(k+1)} \Rightarrow \frac{(k+1)}{(k+2)} = \text{RHS.}$$

as RHS = LHS hence proved.

Eg: Using mathematical induction show that $8^n - 3^n$ is divisible by 5 for $n \geq 1$

$\Rightarrow$ Step 1 for $n = 1$

$$P(n) = 8^n - 3^n, \quad P(1) = 8^1 - 3^1 = 8 - 3 = 5$$

as 5 is divisible by 5.

$\therefore P(1)$ is true.

Step 2 assuming statement is true for $n = k$

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$\therefore P(k) = 8^k - 3^k$ is divisible by 5.

Step 3

we need to prove statement is true for $n = k+1$

i.e. $8^{k+1} - 3^{k+1}$ is divisible by 5.

consider,

$$8^{k+1} - 3^{k+1} = 8^k \cdot 8 - 3^k \cdot 3$$

convert to 8.

$$\Rightarrow 8^k \cdot 8 - 3^k (3 + 5 - 5) \Rightarrow 8^k \cdot 8 - 3^k (8 - 5)$$

$$\Rightarrow 8^k \cdot 8 - 3^k \cdot 8 + 3^k \cdot 5 \Rightarrow 8(8^k - 3^k) + 3^k \cdot 5$$

considering statement from step 2 is true,
and $8^k - 3^k$ is divisible by 5.

Thus

$$\underbrace{8(8^k - 3^k)}_{\substack{\text{divisible} \\ \text{by } 5}} + \underbrace{3^k \cdot 5}_{\substack{\text{divisible} \\ \text{by } 5}}$$

————— X —————

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LOGIC AND PROPOSITIONS

It is declarative sentence which is either true or false but not both.

Logical Connections

- Negation ($\sim$): If P is a statement

P	$\sim P$
T	F
F	T

- Conjunction ($\wedge$, and): The conjunction of statement p and q denoted by $(P \wedge q)$

P	q	$P \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- Disjunction ($\vee$, or): The disjunction of two statement p and q is denoted by $(p \vee q)$

P	q	$P \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

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• **Conditional Statement** ($\rightarrow$, if then) : If

P and q be in any 2 statement then
($p \rightarrow q$) { read as : p implies q }

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

• **Biconditional Statement** ($\leftrightarrow$, if ... only if) :

If p and q be any 2 statements then
($p \leftrightarrow q$) { read as : p if & only if q }.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

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Eg: Translate the following statement into symbolic form.

- 1) Amar and Mohan went up the hill.
- 2) Ram watches television while he has dinner
- 3) Shyam is rich but unhappy.

⇒ 1) p : Amar went up the hill
 q : Mohan went up the hill

Symbolic form: $p \wedge q$

2) p : Ram watches television
 q : Ram has dinner

Symbolic form: $p \wedge q$

3) p : Shyam is rich
 q : Shyam is unhappy

Symbolic form: $p \wedge q$

Q: Write the following sentence in symbolic form

- 1) It is very hot and humid, therefore it will rain
- 2) I am tall but I am not tall.

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- $\Rightarrow$ 1. p : it is very hot
 q : it is very humid
 r : it will rain.

Symbolic form: $(p \wedge q) \rightarrow r$

2. p : I am tall

$\sim p$: I am not tall

Symbolic form: $(p \wedge \sim p)$

Q. Consider the following

~~Anil~~ play

p : Anil is rich

q : Kanchan is poor.

Write each of the following statement in symbolic form.

1. Anil is not rich and Kanchan is poor.

2. It is not true that Anil and Kanchan are both rich.

3. Either Anil is poor or Kanchan is poor.

4. Either Anil or Kanchan is rich.

$\Rightarrow$ 1. $(\sim p \wedge q)$ 2. $\sim (p \wedge \sim q)$

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$$3) (\neg p \vee q)$$

$$4) (p \vee \neg q)$$

Eg: Write truth table of the following

$$1) \neg p \wedge \neg q$$

$$3) (p \vee q) \vee (\neg q)$$

$$2) \neg (\neg p \wedge \neg q)$$

$$4) (\neg p \rightarrow x) \wedge (p \leftrightarrow q)$$

1)

p	q	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

3)

p	q	$p \vee q$	$\neg q$	$(p \vee q) \vee (\neg q)$
T	T	T	F	T
T	F	T	T	T
F	T	T	F	T
F	F	F	T	T

2)

p	q	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	$\neg (\neg p \wedge \neg q)$
T	T	F	F	F	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	T	F

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↙

44	p	q	x	$\neg p$	$\neg p \rightarrow x$	$(p \leftrightarrow q)$	$(\neg p \rightarrow x) \wedge p \leftrightarrow x$
	T	T	T	F	F	T	T
	T	T	F	F	F	T	T
	T	F	T	F	T	F	F
	T	F	F	F	T	F	F
	F	T	T	T	T	F	F
	F	T	F	T	F	F	F
	F	F	T	T	T	T	T
	F	F	F	T	F	T	F

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Q Show that $p \rightarrow (q \rightarrow x)$ and $(p \rightarrow q) \rightarrow (p \rightarrow x)$ are logically equivalent using truth table.

(1)

$\Rightarrow$

p	q	x	$q \rightarrow x$	$p \rightarrow (q \rightarrow x)$
T	T	T	T	T
T	T	F	F	F
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	T
F	F	T	T	T
F	F	F	T	T

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(2)

p	q	r	$p \rightarrow q$	$p \rightarrow r$	$(p \rightarrow q) \rightarrow (p \rightarrow r)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	T	T

From (1) and (2), both statements/expressions
 $p \rightarrow (q \rightarrow r)$ and $(p \rightarrow q) \rightarrow (p \rightarrow r)$
 are equivalent.

Q> show that $p \rightarrow (q \rightarrow r)$ and $p \rightarrow (\neg q \vee r)$
 are logically equivalent using truth table.

p	q	r	$q \rightarrow r$	$p \rightarrow (q \rightarrow r)$
T	T	T	T	T
T	T	F	F	F
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	T
F	F	T	T	T
F	F	F	T	T

(1)

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(2)

p	q	x	$\neg q$	$\neg q \vee x$	$p \rightarrow (\neg q \vee x)$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	T	T	T
F	F	F	T	T	T

from (1) and (2), both statements / expressions
 $p \rightarrow (q \rightarrow x)$ and $p \rightarrow (\neg q \vee x)$
are logically equivalent using truth table.

* Tautology

is a proposition which is always true.

* Contradiction

is a proposition having always false value.

* Contingency

is a proposition which is true for some cases
& false for other cases.

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Q Construct truth tables to determine whether the following is a tautology, contradiction or a contingency.

$$(1) (p \rightarrow q) \leftrightarrow (q \vee \neg p)$$

$$(2) (p \wedge (\neg p \vee q)) \wedge \neg q$$

⇒ (1) ∴ it is a tautology

p	q	$\neg p$	$\neg q$	$(p \rightarrow q)$	$q \vee \neg p$	$(p \rightarrow q) \leftrightarrow (q \vee \neg p)$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

⇒ (2) ∴ it is a contradiction.

p	q	$\neg p$	$\neg q$	$\neg p \vee q$	$p \wedge (\neg p \vee q)$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	F
F	F	T	T	T	F

↳ continued

$(p \wedge (\neg p \vee q)) \wedge \neg q$
F
F
F
F

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Q: A security system sounds an alarm (P) only when motion is detected and a door or a window is open. Build the truth table to verify all combinations under which the alarm should trigger.

⇒ (P) only when motion is detected AND (a door OR window is opened)

p : motion is detected

q : door is opened

r : window is opened.

$$\text{Alarm} = p \overset{\text{and}}{\wedge} (q \overset{\text{or}}{\vee} r).$$

p	q	r	$q \vee r$	$p \wedge (q \vee r)$	
T	T	T	T	T	} tautology.
T	T	F	T	T	
T	F	T	T	T	
T	F	F	F	F	} alarm will not ring
F	T	T	T	F	
F	T	F	T	F	
F	F	T	T	F	
F	F	F	F	F	

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The alarm triggers in 3 combination:

1) motion detected + door open + window open

2) motion detected + door open

3) motion detected + window open

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* CONJUNCTIVE NORMAL FORM (CNF)

A compound statement is a conjunctive normal form if it is obtained by operating AND among variables connected with OR's

AND

Eg> (1) $(p \vee q) \wedge (q \vee r)$

(2) $(\neg p \vee q \vee r) \wedge (s \vee r)$

* DISJUNCTIVE NORMAL FORM (DNF)

A compound statement is a disjunctive normal form if it is obtained by operating OR among variables connected with AND

Eg> (1) $(p \wedge q) \underset{\text{OR}}{\vee} (\neg p \wedge q) \underset{\text{OR}}{\vee} (p \wedge \neg q) \underset{\text{OR}}{\vee} (\neg r)$

Eg> Obtain CNF of $p \wedge (p \rightarrow q)$
→ forming truth table

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H
C
v
R

	p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$
1	T	T	T	T
2	T	F	F	F
3	F	T	T	F
4	F	F	T	F

→ Rows = 2, 3, 4

for CNF, identify rows where the formula is false.

The formula is False in row 2, 3, 4.

Rule FOR CNF

1> if variable is true, take negation ($\neg$)

2> if variable is false, take variable itself.

Row 2 → ($p = T, q = F$) = ($\neg p \vee q$)

Row 3 → ($p = F, q = T$) = ($p \vee \neg q$)

Row 4 → ($p = F, q = F$) = ($p \vee q$)

combine with AND,

CNF = ($\neg p \vee q$) $\wedge$ ($p \vee \neg q$) $\wedge$ ($p \vee q$)

Eg> Obtain CNF of $(\neg p \rightarrow r) \wedge (p \wedge q)$
→ forming truth table.

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p	q	r	$\neg p$	$\neg p \rightarrow r$	$p \wedge q$	$(\neg p \rightarrow r) \wedge (p \wedge q)$	
T	T	T	F	T	T	T	1
T	T	F	F	T	T	T	2
T	F	T	F	T	F	F	3
T	F	F	F	T	F	F	4
F	T	T	T	T	F	F	5
F	T	F	T	F	F	F	6
F	F	T	T	T	F	F	7
F	F	F	T	F	F	F	8

for CNF, identify rows where the formula is false.

The formula is false in row 3, 4, 5, 6, 7, 8.

$$\text{Row 3} = (T, F, T) = (\neg p \vee q \vee \neg r)$$

$$\text{Row 4} = (T, F, F) = (\neg p \vee q \vee r)$$

$$\text{Row 5} = (F, T, T) = (p \vee \neg q \vee \neg r)$$

$$\text{Row 6} = (F, T, F) = (p \vee \neg q \vee r)$$

$$\text{Row 7} = (F, F, T) = (\neg p \vee q \vee \neg r)$$

$$\text{Row 8} = (F, F, F) = (\neg p \vee q \vee r)$$

combine with AND,

$$\text{CNF} = (\neg p \vee q \vee \neg r) \wedge (\neg p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge (p \vee \neg q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (\neg p \vee q \vee r)$$

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Eg> Obtain DNF of $(p \rightarrow q) \wedge (\neg p \wedge q)$
→ forming truth table

p	q	$p \rightarrow q$	$\neg p$	$\neg p \wedge q$	$(p \rightarrow q) \wedge (\neg p \wedge q)$	
T	T	T	F	F	F	1
T	F	F	F	F	F	2
F	T	T	T	T	T	3
F	F	T	T	F	F	4

for DNF, identify row where the Formula is True.

The Formula is True in Row 3.

Rule for DNF

1> If variable is True, then take Variable itself.

2> If variable is False, take its negation

Row 3 = (F, T) = $(\neg p \wedge q)$

Combine with OR,

DNF = $(\neg p \wedge q)$.

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Eg) obtain DNF of $(p \wedge (p \rightarrow q)) \rightarrow q$
forming truth table

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$p \wedge (p \rightarrow q) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

$$\text{Row 1} = (T, T) = (p \wedge q)$$

$$\text{Row 2} = (T, F) = (p \wedge \neg q)$$

$$\text{Row 3} = (F, T) = (\neg p \wedge q)$$

$$\text{Row 4} = (F, F) = (\neg p \wedge \neg q)$$

combine with OR

$$\text{DNF} = (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$$

Eg) obtain DNF of $\neg (p \rightarrow (q \wedge r))$
forming truth table $\rightarrow$

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and imply ($TF \rightarrow F$)

Row 4 = (T, F, F) = $(p \wedge \neg q, \wedge \neg r)$

$$\vee (p \wedge \neg q \wedge \neg r)$$

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QUANTIFIERS

* Universal Quantifiers ($\forall$)

True for every value of the specific variable ($\forall$)

$\forall x P(x)$ read as "for every value of x , $P(x)$ is true."

* Existential Quantifiers ($\exists$)

True for some value of the specific variable ($\exists$)

$\exists x P(x)$ read as "for some value of x , $P(x)$ is true?"

* Predicate

A part of declarative sentence describing the properties of an object.

* Negation of Quantifiers

$$\neg (\overset{\text{for every}}{\forall x P(x)}) \Leftrightarrow \exists x \overset{\text{for some}}{[\neg P(x)]}$$

$$\neg [\exists x P(x)] \Leftrightarrow \forall x [\neg P(x)]$$

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Eg> Let $A = \{1, 2, 3, 4, \dots, \dots, \dots, 10\}$

Consider each of the following sentences, if it is a statement, then determine its truth value. If it is propositional function, then determine its truth set.

$$(i) (\forall x \in A) (\exists y \in A) (x + y < 14)$$

for every x (belong to A) we need some y (belong to A) such that it satisfies $(x + y < 14)$

$\Rightarrow$ for every x we need at least 1 y such that
 $x + y < 14$

choose $y = 1$, (maximum) $x = 10$

$$\therefore 10 + 1 = 11 < 14$$

and thus this statement is true.

$$(ii) (\forall x \in A) (\forall y \in A) (x + y < 14)$$

$\Rightarrow$ for every x we need every y such that
 $x + y < 14$

choose $y = 10$, $x = 10$

$$\therefore 10 + 10 = 20 > 14 \text{ or } 20 \not< 14$$

and thus this statement is false.

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Propositional function

$$(iii) \quad (\forall y \in A) \quad (x + y < 14) \quad A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

here x is free so it is Propositional function
for every y (belongs to A) $\Rightarrow$ consider largest value.

$$x + 10 < 14$$

$$x < 14 - 10$$

$$\boxed{x < 4}$$

$$\text{Truth set} = \{1, 2, 3\}$$

$$(iv) \quad (\exists y \in A) \quad (x + y < 14)$$

we need at least 1 value of y belongs to A

$$\therefore \text{consider } y = 1, \quad x + 1 < 14$$

$$\Rightarrow x < 14 - 1$$

$$\Rightarrow \boxed{x < 13}$$

$$\text{Truth set} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

Eg> For the universe of all integers, let $P(x)$, $Q(x)$, $R(x)$, $S(x)$ and $T(x)$ be the following statements.

$P(x)$: $x > 0$, $R(x)$: x is a perfect square

$Q(x)$: x is even $S(x)$: x is divisible by 4

$T(x)$: x is divisible by 5.

Write following statement in symbolic form

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(i) at least one integer is even $\rightarrow \exists x Q(x)$

(ii) There exist a positive integer that is even $\rightarrow$
 $\exists x [P(x) \wedge Q(x)]$

(iii) There exists an even integer divisible by 5.
 $\exists x [Q(x) \wedge T(x)]$

(iv) If x is Even and x is a perfect square then x is divisible by 4.

$$\forall x [(Q(x) \wedge R(x)) \rightarrow S(x)]$$

Eg: Negate each of the following in such a way that negation does not appear before the quantifier.

(i) $\exists x \forall y [x > y]$

(ii) $\forall x \exists y [x - y = 1]$

(iii) $\forall x \forall y [(y > 0) \rightarrow (xy > 0)]$

$\Rightarrow$ (i) $\neg [\exists x \forall y [x > y]]$

$\equiv \forall x \exists y \neg (x > y)$

$\equiv \forall x \exists y (x \leq y)$

(ii) $\neg [\forall x \exists y [x - y = 1]]$

$\equiv \exists x \forall y \neg (x - y = 1)$

$\equiv \exists x \forall y (x - y \neq 1)$

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$$(iii) \neg [\forall x \forall y [(y > 0) \rightarrow (xy > 0)]]$$

$$\equiv \exists x \exists y \neg \underbrace{(y > 0)}_p \rightarrow \underbrace{(xy > 0)}_q$$

use Rule

$$\neg p \rightarrow q \Rightarrow p \wedge \neg q$$

$$\equiv \exists x \exists y ((y > 0) \wedge \neg (xy > 0))$$

$$\equiv \exists x \exists y ((y > 0) \wedge (xy \leq 0))$$

— X —

~~C~~
~~✱~~